Chapter 1. The Algebra and Topology of $\mathbb{R}^n$

Linear algebra, matrix inversion and determinants, metric topology, compactness and connectedness in $\mathbb{R}^n$

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Overview

This chapter collects the linear algebra and the topology of $\mathbb{R}^n$ used throughout. Sections 1 and 3 are review; Sections 2 and 4 contain material that may be new (elementary matrices, the axiomatic determinant, the ϵ-neighborhood theorem, compactness of rectangles). Results that Munkres states without proof are given short proofs here, marked *Proof (supplied).* Elements of $\mathbb{R}^n$ are written as column matrices unless stated otherwise.

§1. Review of Linear Algebra

1.1 Vector spaces

Definition 1.1 (vector space)

A set V with an operation $\mathbf{x}+\mathbf{y}$ (vector addition) and an operation $c\mathbf{x}$ (scalar multiplication by $c \in \mathbb{R}$) is a **vector space** (or **linear space**) if for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$ and $c, d \in \mathbb{R}$:

(1) $\mathbf{x}+\mathbf{y} = \mathbf{y}+\mathbf{x}$;

(2) $\mathbf{x}+(\mathbf{y}+\mathbf{z}) = (\mathbf{x}+\mathbf{y})+\mathbf{z}$;

(3) there is a unique vector $\mathbf{0}$ with $\mathbf{x}+\mathbf{0} = \mathbf{x}$ for all $\mathbf{x}$;

(4) $\mathbf{x}+(-1)\mathbf{x} = \mathbf{0}$;

(5) $1\mathbf{x} = \mathbf{x}$;

(6) $c(d\mathbf{x}) = (cd)\mathbf{x}$;

(7) $(c+d)\mathbf{x} = c\mathbf{x}+d\mathbf{x}$;

(8) $c(\mathbf{x}+\mathbf{y}) = c\mathbf{x}+c\mathbf{y}$.

The basic example is $\mathbb{R}^n$, the set of n-tuples $\mathbf{x} = (x_1, \dots, x_n)$ with componentwise operations $\mathbf{x}+\mathbf{y} = (x_1+y_1, \dots, x_n+y_n)$, $c\mathbf{x} = (cx_1, \dots, cx_n)$.

A subset $W \subset V$ is a **linear subspace** if $\mathbf{x}+\mathbf{y} \in W$ and $c\mathbf{x} \in W$ whenever $\mathbf{x}, \mathbf{y} \in W$, $c \in \mathbb{R}$; then W is a vector space in its own right. In the first part of the book only $\mathbb{R}^n$ and its subspaces occur.

Definition 1.2 (span, independence, basis, dimension)

Let $\mathbf{a}_1, \dots, \mathbf{a}_m \in V$.

- They **span** V if every $\mathbf{x} \in V$ can be written in *at least* one way as a **linear combination** $\mathbf{x} = c_1\mathbf{a}_1 + \cdots + c_m\mathbf{a}_m$.
- They are **independent** if every $\mathbf{x} \in V$ can be written in *at most* one way as such a combination; equivalently, $d_1\mathbf{a}_1 + \cdots + d_m\mathbf{a}_m = \mathbf{0}$ forces $d_1 = \cdots = d_m = 0$.
- They form a **basis** if they both span V and are independent.

If V has a basis of m vectors, m is the **dimension** of V (by Theorem 1.1 this is well defined); the zero space has dimension 0.

Theorem 1.1

Suppose V has a basis consisting of m vectors. Then any set of vectors that spans V has at least m vectors, and any set of vectors of V that is independent has at most m vectors. In particular, any basis for V has exactly m vectors.

Proof (supplied). It suffices to show: if $\mathbf{w}_1,\dots,\mathbf{w}_k$ span V and $\mathbf{v}_1,\dots,\mathbf{v}_l$ are independent, then $l \le k$ (apply it with the basis as the spanning set, and with the basis as the independent set). Write $\mathbf{v}_j = \sum_{i=1}^{k} a_{ij}\mathbf{w}_i$. If $l > k$, the homogeneous linear system $\sum_{j=1}^{l} a_{ij}x_j = 0$ $(i = 1,\dots,k)$ has more unknowns than equations, hence (by Gauss–Jordan reduction: an echelon form has at most k corner entries, so some unknown is free) a nonzero solution $(x_1,\dots,x_l)$. Then $\sum_j x_j\mathbf{v}_j = \sum_i(\sum_j a_{ij}x_j)\mathbf{w}_i = \mathbf{0}$, contradicting independence. ■

$\mathbb{R}^n$ has dimension n: the **standard basis** is $\mathbf{e}_1 = (1,0,\dots,0)$, $\mathbf{e}_2 = (0,1,0,\dots,0)$, $\dots$, $\mathbf{e}_n = (0,\dots,0,1)$. Every basis of $\mathbb{R}^n$ has exactly n vectors. (Spanning, independence and bases make sense for infinite sets of vectors too, but we shall not need this.)

Theorem 1.2

Let V be a vector space of dimension m. If W is a linear subspace of V different from V, then W has dimension less than m. Furthermore, any basis $\mathbf{a}_1,\dots,\mathbf{a}_k$ for W may be extended to a basis $\mathbf{a}_1,\dots,\mathbf{a}_k,\mathbf{a}_{k+1},\dots,\mathbf{a}_m$ for V.

Proof (supplied). First, W has a finite basis: choose independent vectors of W one at a time as long as possible; by Theorem 1.1 the process stops after at most m steps, and a maximal independent subset of W spans W (if $\mathbf{x} \in W$ were not a combination of them, adjoining $\mathbf{x}$ would keep independence). Now let $\mathbf{a}_1,\dots,\mathbf{a}_k$ be a basis of W. If they do not span V, pick $\mathbf{a}_{k+1} \in V$ not in their span; then $\mathbf{a}_1,\dots,\mathbf{a}_{k+1}$ are independent (a relation with nonzero coefficient of $\mathbf{a}_{k+1}$ would put $\mathbf{a}_{k+1}$ in the span; one with zero coefficient contradicts independence of the $\mathbf{a}_i$). Repeat; by Theorem 1.1 the process stops when we have an independent spanning set of m vectors. If $W \neq V$, at least one vector was added, so $k < m$. ■

1.2 Inner products and norms

Definition 1.3 (inner product)

An **inner product** on V assigns to $\mathbf{x},\mathbf{y} \in V$ a real number $\langle \mathbf{x},\mathbf{y}\rangle$ such that for all $\mathbf{x},\mathbf{y},\mathbf{z}$ and scalars c:
(1) $\langle \mathbf{x},\mathbf{y}\rangle = \langle \mathbf{y},\mathbf{x}\rangle$;
(2) $\langle \mathbf{x}+\mathbf{y},\mathbf{z}\rangle = \langle \mathbf{x},\mathbf{z}\rangle + \langle \mathbf{y},\mathbf{z}\rangle$;
(3) $\langle c\mathbf{x},\mathbf{y}\rangle = c\langle \mathbf{x},\mathbf{y}\rangle = \langle \mathbf{x},c\mathbf{y}\rangle$;
(4) $\langle \mathbf{x},\mathbf{x}\rangle > 0$ if $\mathbf{x} \neq \mathbf{0}$. V together with an inner product is an **inner product space**.

On $\mathbb{R}^n$ we use the **dot product** $\langle \mathbf{x},\mathbf{y}\rangle = x_1y_1 + \cdots + x_ny_n$ (written $\langle \mathbf{x},\mathbf{y}\rangle$ rather than $\mathbf{x}\cdot\mathbf{y}$ to avoid confusion with the matrix product).

Definition 1.4 (norm)

The **length** (or norm) in an inner product space is $\|\mathbf{x}\| = \langle \mathbf{x},\mathbf{x}\rangle^{1/2}$. It satisfies (1) $\|\mathbf{x}\| > 0$ if $\mathbf{x} \neq \mathbf{0}$;
(2) $\|c\mathbf{x}\| = |c|\,\|\mathbf{x}\|$;
(3) $\|\mathbf{x}+\mathbf{y}\| \le \|\mathbf{x}\| + \|\mathbf{y}\|$ (**triangle inequality**); equivalently (3′) $\|\mathbf{x}-\mathbf{y}\| \ge \|\mathbf{x}\| - \|\mathbf{y}\|$. Any function $V \to \mathbb{R}$ with properties (1)–(3) is called a **norm** on V.

Proof of (3) (supplied). *Cauchy–Schwarz*: $|\langle \mathbf{x},\mathbf{y}\rangle| \le \|\mathbf{x}\|\|\mathbf{y}\|$. This is trivial if $\mathbf{x}$ or $\mathbf{y}$ is $\mathbf{0}$; otherwise put $c = 1/\|\mathbf{x}\|$, $d = 1/\|\mathbf{y}\|$; then $0 \le \|c\mathbf{x} \pm d\mathbf{y}\|^2 = 2 \pm 2cd\langle \mathbf{x},\mathbf{y}\rangle$, so $|\langle \mathbf{x},\mathbf{y}\rangle| \le 1/(cd)$. Hence $\|\mathbf{x}+\mathbf{y}\|^2 = \|\mathbf{x}\|^2 + 2\langle \mathbf{x},\mathbf{y}\rangle + \|\mathbf{y}\|^2 \le (\|\mathbf{x}\| + \|\mathbf{y}\|)^2$. Applying (3) to $\mathbf{x} = (\mathbf{x}-\mathbf{y}) + \mathbf{y}$ gives (3′). ■

On $\mathbb{R}^n$ we have the **euclidean norm** $\|\mathbf{x}\|$ coming from the dot product, and the **sup norm**

$$|\mathbf{x}| = \max\{|x_1|, \dots, |x_n|\}.$$

The sup norm is often more convenient. The two are related by

$$|\mathbf{x}| \le \|\mathbf{x}\| \le \sqrt{n}\,|\mathbf{x}|, \tag{1.1}$$

since $\max_i x_i^2 \le \sum_i x_i^2 \le n \max_i x_i^2$.

1.3 Matrices

An n by m matrix A has entries a_{ij} (i = row index, j = column index). With entrywise addition and scalar multiplication the n by m matrices form a vector space (essentially $\mathbb{R}^{nm}$). If A is n by m and B is m by p, the **product** $C = A \cdot B$ is the n by p matrix with

$$c_{ij} = \sum_{k=1}^{m} a_{ik} b_{kj}.$$

It satisfies $A \cdot (B \cdot C) = (A \cdot B) \cdot C$, $A \cdot (B + C) = A \cdot B + A \cdot C$, $(A + B) \cdot C = A \cdot C + B \cdot C$, $(cA) \cdot B = c(A \cdot B) = A \cdot (cB)$, and $I_n \cdot A = A = A \cdot I_m$ for A of size n by m, where the **identity matrix** I_k has entries δ_{ij} ($\delta_{ij} = 1$ if $i = j$, 0 otherwise).

The sup norm extends to matrices: $|A| = \max\{|a_{ij}| : i = 1, \dots, n,\ j = 1, \dots, m\}$; it is a norm.

Theorem 1.3

If A has size n by m and B has size m by p, then $|A \cdot B| \le m|A|\,|B|$.

Proof (supplied). $|c_{ij}| = |\sum_{k=1}^m a_{ik}b_{kj}| \le \sum_{k=1}^m |A||B| = m|A||B|$ for every i, j. ■

1.4 Linear transformations

Definition 1.5 (linear transformation)

A function $T : V \to W$ between vector spaces is a **linear transformation** if $T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y})$ and $T(c\mathbf{x}) = cT(\mathbf{x})$. If moreover T is a bijection, T is a **linear isomorphism**.

Composites of linear transformations are linear, and the inverse of a linear isomorphism is linear.

Theorem 1.4

Let V be a vector space with basis $\mathbf{a}_1, \dots, \mathbf{a}_m$. Let W be a vector space. Given any m vectors $\mathbf{b}_1, \dots, \mathbf{b}_m$ in W, there is exactly one linear transformation $T : V \to W$ such that $T(\mathbf{a}_i) = \mathbf{b}_i$ for all i.

Proof (supplied). Each $\mathbf{x} \in V$ has a unique expression $\mathbf{x} = \sum c_i \mathbf{a}_i$; define $T(\mathbf{x}) = \sum c_i \mathbf{b}_i$. Uniqueness of the coefficients makes T well defined, and since the coefficients of $\mathbf{x} + \mathbf{y}$ and $c\mathbf{x}$ are the sums and multiples of those of $\mathbf{x}, \mathbf{y}$, T is linear. Any linear T' with $T'(\mathbf{a}_i) = \mathbf{b}_i$ satisfies $T'(\sum c_i \mathbf{a}_i) = \sum c_i \mathbf{b}_i$, so $T' = T$. ■

The maps $(x_1, \dots, x_n) \mapsto [x_1 \ \cdots \ x_n]$ (**row matrix**) and $(x_1, \dots, x_n) \mapsto$ the n by 1 **column matrix** with entries x_i are linear isomorphisms. Representing elements of $\mathbb{R}^m$ as column matrices, a fixed n by m matrix A defines a linear map $T : \mathbb{R}^m \to \mathbb{R}^n$, $T(\mathbf{x}) = A \cdot \mathbf{x}$. Conversely every linear $T : \mathbb{R}^m \to \mathbb{R}^n$ has this form: let $\mathbf{b}_j = T(\mathbf{e}_j)$ and $A = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_m]$; then $A \cdot \mathbf{e}_j = \mathbf{b}_j = T(\mathbf{e}_j)$ for all j, so $A \cdot \mathbf{x} = T(\mathbf{x})$ by Theorem 1.4.

Convention

Throughout, the elements of $\mathbb{R}^n$ are represented by column matrices, unless specifically stated otherwise.

1.5 Rank of a matrix

For an n by m matrix A, the **column space** is the subspace of $\mathbb{R}^n$ spanned by the columns; its dimension is the **column rank** ($\leq \min(m, n)$). The **row space** is the subspace of $\mathbb{R}^m$ spanned by the rows; its dimension is the **row rank**.

Theorem 1.5

For any matrix A, the row rank of A equals the column rank of A.

Proof (supplied). Let r be the column rank and let $\mathbf{c}_1, \ldots, \mathbf{c}_r \in \mathbb{R}^n$ be a basis of the column space; let $C = [\mathbf{c}_1 \ \cdots \ \mathbf{c}_r]$ (n by r). Each column of A is a combination of the $\mathbf{c}_k$, so $A = C \cdot D$ for some r by m matrix D. Then each row of A is a combination of the r rows of D (row i of A is $\sum_k c_{ik}\cdot$(row k of D)), so the row rank of A is $\leq r$. Applying this to A^{tr} (whose columns are the rows of A) gives the reverse inequality. ■

The common value is the **rank** of A.

Elementary row operations on a matrix are:

(1) exchange rows $i_1 \neq i_2$;

(2) replace row i_1 by itself plus c times row i_2 ($i_1 \neq i_2$);

(3) multiply row i by a scalar $\lambda \neq 0$. Each is invertible, and its inverse is an elementary operation of the same type.

Theorem 1.6

If B is the matrix obtained by applying an elementary row operation to A, then $\operatorname{rank} B = \operatorname{rank} A$.

Proof (supplied). Each row of B is a linear combination of rows of A, so the row space of B is contained in that of A. Since A is obtained from B by the inverse elementary operation, the reverse inclusion holds too. The row spaces coincide, so the row ranks agree. ■

Gauss–Jordan reduction applies elementary operations to bring A to **echelon form** (stairstep form): below the "stairsteps" all entries are 0 and the corner entries (pivots) are nonzero. Operations of types (1) and (2) suffice. The nonzero rows of a matrix in echelon form are independent, so its rank is the number of nonzero rows. Using further operations of types (2) and (3) one reaches the **reduced echelon form**, in which each corner entry is 1 and all entries directly above it are 0.

1.6 Transpose

The **transpose** of the n by m matrix A is the m by n matrix A^{tr} with entries $d_{ij} = a_{ji}$. Then $(A^{\mathrm{tr}})^{\mathrm{tr}} = A$, $(A + B)^{\mathrm{tr}} = A^{\mathrm{tr}} + B^{\mathrm{tr}}$, $(A \cdot C)^{\mathrm{tr}} = C^{\mathrm{tr}} \cdot A^{\mathrm{tr}}$, and $\operatorname{rank} A^{\mathrm{tr}} = \operatorname{rank} A$ (the row rank of A^{tr} is the column rank of A).

§2. Matrix Inversion and Determinants

2.1 Elementary matrices

Definition 2.1 (elementary matrix)

An **elementary matrix** of size n by n is the matrix obtained by applying one of the elementary row operations to I_n.

There are three types: E (rows i_1, i_2 of I_n exchanged), E' (I_n with an extra entry c in row i_1, column i_2), and E'' (I_n with the (i,i) entry replaced by $\lambda \neq 0$).

Theorem 2.1

Let A be an n by m matrix. Any elementary row operation on A may be carried out by premultiplying A by the corresponding elementary matrix.

Proof. Direct computation: multiplying A on the left by E interchanges rows i_1 and i_2 of A; by E' replaces row i_1 by itself plus c times row i_2; by E'' multiplies row i by λ. ■

2.2 The inverse of a matrix

Definition 2.2 (left and right inverse)

Let A be n by m, and B, C be m by n. B is a **left inverse** for A if $B \cdot A = I_m$; C is a **right inverse** for A if $A \cdot C = I_n$.

Theorem 2.2

If A has both a left inverse B and a right inverse C, then they are unique and equal.

Proof. $C = I_m \cdot C = (B \cdot A) \cdot C = B \cdot (A \cdot C) = B \cdot I_n = B$. If B_1 is another left inverse, the same computation gives $C = B_1$, so $B_1 = B$; similarly C is unique. ■

Definition 2.3 (invertible)

If A has both a left and a right inverse, A is **invertible**; the unique two-sided inverse is denoted A^{-1}.

Theorem 2.3

Let A be a matrix of size n by m. If A is invertible, then $n = m = \operatorname{rank} A$.

Proof.

Step 1. For any k by n matrix D, $\operatorname{rank}(D \cdot A) \leq \operatorname{rank} A$. Indeed, if R is a 1 by n row matrix, $R \cdot A$ is a linear combination of the rows of A. Each row of $D \cdot A$ is (row of D)$\cdot A$, so the row space of $D \cdot A$ lies in the row space of A.

Step 2. If A has a left inverse B, then $\operatorname{rank} A = m$ (the number of columns). Indeed $m = \operatorname{rank} I_m = \operatorname{rank}(B \cdot A) \leq \operatorname{rank} A$ by Step 1, while the row space of A is a subspace of $\mathbb{R}^m$, so $\operatorname{rank} A \leq m$.

Step 3. Let $B = A^{-1}$. Since B is a left inverse, $\operatorname{rank} A = m$ by Step 2. Since B is a right inverse, $B^{\mathrm{tr}} \cdot A^{\mathrm{tr}} = I_n^{\mathrm{tr}} = I_n$, so B^{tr} is a left inverse of the m by n matrix A^{tr} and Step 2 gives $\operatorname{rank} A = \operatorname{rank} A^{\mathrm{tr}} = n$. ■

Theorem 2.4

Let A be a matrix of size n by m with $n = m = \operatorname{rank} A$. Then A is invertible; furthermore, A equals a product of elementary matrices.

Proof.

Step 1. Every elementary matrix is invertible with elementary inverse: E is its own inverse, E' with c replaced by $-c$ inverts E', and E'' with λ replaced by $1/\lambda$ inverts E''.

Step 2. Reduce A to reduced echelon form C by elementary operations. C is square of rank n, so it has n nonzero rows; hence $C = I_n$. By Theorem 2.1 there are elementary matrices $E_1, \dots, E_k$ with $E_k \cdots E_2 E_1 A = I_n$. Multiplying on the left successively by $E_k^{-1}, \dots, E_1^{-1}$ gives

$$A = E_1^{-1} E_2^{-1} \cdots E_k^{-1},$$

a product of elementary matrices, and $B = E_k E_{k-1} \cdots E_1$ is both a left and a right inverse for A. ■

Theorem 2.5

If A is a square matrix and B is a left inverse for A, then B is also a right inverse for A.

Proof. By Step 2 of the proof of Theorem 2.3, $\operatorname{rank} A$ equals the number of columns, which (as A is square) equals the number of rows. By Theorem 2.4, A has an inverse, which equals B by Theorem 2.2. ■

An n by n matrix is **singular** if $\operatorname{rank} A < n$, **non-singular** otherwise. By the above, A is invertible iff it is non-singular.

2.3 Determinants

Since $|A|$ denotes the sup norm, the determinant is written $\det A$. We characterize it by axioms; the existence of such a function for all n is proved in Chapter 6 (§27).

Definition 2.4 (determinant function)

A function assigning to each n by n matrix A a real number $\det A$ is a **determinant function** if:

(1) if B is obtained from A by exchanging two rows, then $\det B = -\det A$;

(2) for each i, $\det A$ is linear as a function of the ith row alone: if $A_i(\mathbf{x})$ denotes A with row i replaced by $\mathbf{x}$, then $\det A_i(a\mathbf{x} + b\mathbf{y}) = a \det A_i(\mathbf{x}) + b \det A_i(\mathbf{y})$;

(3) $\det I_n = 1$.

Example 2.1

For 1 by 1 matrices $\det[a] = a$; for 2 by 2 matrices $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$; for 3 by 3,

$$\det \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} = a_1 b_2 c_3 + a_2 b_3 c_1 + a_3 b_1 c_2 - a_3 b_2 c_1 - a_1 b_3 c_2 - a_2 b_1 c_3.$$

These satisfy the axioms. For larger matrices a direct formula has $n!$ terms (24 for $n = 4$, 120 for $n = 5$), so a less direct approach is needed.

Theorem 2.6

Let A be an n by n matrix.

(a) If E is the elementary matrix exchanging rows i_1 and i_2, then $\det(E \cdot A) = -\det A$.

(b) If E' replaces row i_1 of A by itself plus c times row i_2, then $\det(E' \cdot A) = \det A$.

(c) If E'' multiplies row i of A by $\lambda \neq 0$, then $\det(E'' \cdot A) = \lambda \det A$.

(d) If $A = I_n$, then $\det A = 1$.

Proof. (a) is Axiom 1 and (d) is Axiom 3;

(c) is the special case $\det A_i(\lambda \mathbf{x}) = \lambda \det A_i(\mathbf{x})$ of Axiom 2. For (b): if A has two equal rows, exchanging them does not change A but changes the sign of $\det A$, so $\det A = 0$. Let $i = i_1$, $\mathbf{x} =$ row i_1, $\mathbf{y} =$ row i_2. Then

$$\det(E' \cdot A) = \det A_i(\mathbf{x} + c\mathbf{y}) = \det A_i(\mathbf{x}) + c \det A_i(\mathbf{y}) = \det A_i(\mathbf{x}) = \det A,$$

since $A_i(\mathbf{y})$ has two equal rows. ■

Taking $A = I_n$: $\det E = -1$, $\det E' = 1$, $\det E'' = \lambda$.

Theorem 2.7

Let A be a square matrix. If the rows of A are independent, then $\det A \neq 0$; if the rows are dependent, then $\det A = 0$. Thus an n by n matrix A has rank n if and only if $\det A \neq 0$.

Proof. If row i of A is zero, then $\det A = 0$: multiplying that row by 2 leaves A unchanged but doubles $\det A$. An elementary operation multiplies the determinant by -1, 1 or $\lambda \neq 0$, so it does not affect whether $\det A$ vanishes. Reduce A to echelon form B (types (1), (2)). If the rows of A are dependent, $\operatorname{rank} B = \operatorname{rank} A < n$, so B has a zero row, $\det B = 0$, and $\det A = 0$. If they are independent, reduce further to reduced echelon form C; being square of rank n, $C = I_n$, so $\det C = 1 \neq 0$ and $\det A \neq 0$. ■

Theorem 2.8

Given a square matrix A, reduce it to echelon form B by elementary row operations of types (1) and (2). If B has a zero row, then $\det A = 0$. Otherwise, let k be the number of row exchanges involved; then $\det A = (-1)^k$ times the product of the diagonal entries of B.

Proof. If B has a zero row, $\operatorname{rank} A < n$ and $\det A = 0$. Otherwise $\det A = (-1)^k \det B$ by Theorem 2.6(a),(b), and B is upper triangular with nonzero diagonal entries $b_{11}, \ldots, b_{nn}$. Using type (2) operations, clear the entries above the diagonal; the diagonal is unaffected, giving a diagonal matrix C with $\det C = \det B$. Multiplying row i of C by $1/b_{ii}$ for each i yields I_n, so by Theorem 2.6(c),(d), $1 = \det I_n = (1/b_{11}) \cdots (1/b_{nn}) \det C$. Hence $\det B = \det C = b_{11} \cdots b_{nn}$. ■

Corollary 2.9

The determinant function is uniquely characterized by its three axioms. It is also characterized by the four properties listed in Theorem 2.6.

Proof. The calculation of $\det A$ in Theorem 2.8 uses only properties (a)–(d) of Theorem 2.6, which follow from the axioms. ■

Theorem 2.10

Let A and B be n by n matrices. Then $\det(A \cdot B) = (\det A) \cdot (\det B)$.

Proof.

Step 1. The theorem holds when A is elementary: $\det(E \cdot B) = -\det B = (\det E)(\det B)$, $\det(E' \cdot B) = \det B = (\det E')(\det B)$, $\det(E'' \cdot B) = \lambda \det B = (\det E'')(\det B)$.

Step 2. If $\operatorname{rank} A = n$, $A = E_1 \cdots E_k$ is a product of elementary matrices (Theorem 2.4), and repeated use of Step 1 gives $\det(A \cdot B) = (\det E_1) \cdots (\det E_k)(\det B)$ for all B. With $B = I_n$ this reads $\det A = (\det E_1) \cdots (\det E_k)$, and the theorem follows.

Step 3. If $\operatorname{rank} A < n$, then $\operatorname{rank}(A \cdot B) = \operatorname{rank}(B^{\mathrm{tr}} \cdot A^{\mathrm{tr}}) \leq \operatorname{rank} A^{\mathrm{tr}} < n$ by Step 1 of Theorem 2.3, so both sides vanish by Theorem 2.7. ■

Theorem 2.11

$\det A^{\mathrm{tr}} = \det A$.

Proof.

Step 1. For elementary matrices: $E^{\mathrm{tr}} = E$, $(E'')^{\mathrm{tr}} = E''$, and $(E')^{\mathrm{tr}}$ is again of type (2), so both have determinant 1.

Step 2. If $\operatorname{rank} A = n$, write $A = E_1 \cdots E_k$. Then by Theorem 2.10 and Step 1,

$$\det A^{\mathrm{tr}} = \det(E_k^{\mathrm{tr}} \cdots E_1^{\mathrm{tr}}) = (\det E_k^{\mathrm{tr}}) \cdots (\det E_1^{\mathrm{tr}}) = (\det E_1) \cdots (\det E_k) = \det A.$$

Step 3. If $\operatorname{rank} A < n$, then $\operatorname{rank} A^{\mathrm{tr}} < n$ and both determinants are 0. ■

2.4 A formula for A^{-1}

Definition 2.5 (minor, cofactor)

For an n by n matrix A, the $(n-1)$ by $(n-1)$ matrix A_{ij} obtained by deleting row i and column j is the (i,j)**-minor**; the number $(-1)^{i+j} \det A_{ij}$ is the (i,j)**-cofactor** of A.

Lemma 2.12

Let A be an n by n matrix and let b denote its entry in row i and column j.

(a) If all the entries in row i other than b vanish, then $\det A = b(-1)^{i+j} \det A_{ij}$.

(b) The same holds if all the entries in column j other than b vanish.

Proof.

Step 1. Fix $b, a_2, \ldots, a_n$. For an $(n-1)$ by $(n-1)$ matrix D let $A(D)$ be the n by n matrix with first row $(b, a_2, \ldots, a_n)$, first column $(b, 0, \ldots, 0)^{\mathrm{tr}}$ and lower right block D. We show $\det A(D) = b \det D$. If $b = 0$, $\operatorname{rank} A(D) < n$ and both sides vanish. If $b \neq 0$, put $f(D) = (1/b) \det A(D)$. Exchanging two rows of D, adding a multiple of one row of D to another, or multiplying a row of D by λ has the same effect on the corresponding rows $(i+1)$ of $A(D)$, so f changes by -1, 1, λ, respectively. If $D = I_{n-1}$, $A(D)$ is in echelon form and $\det A(D) = b$ by Theorem 2.8, so $f(I_{n-1}) = 1$. By Corollary 2.9, $f(D) = \det D$.

Step 2. Taking transposes (Theorem 2.11), the same formula holds for the matrix with first row $(b, 0, \ldots, 0)$, first column $(b, a_2, \ldots, a_n)^{\mathrm{tr}}$ and block D.

Step 3. In the general case, $i-1$ exchanges of adjacent rows bring row i to the top without changing the order of the other rows; then $j-1$ exchanges of adjacent columns bring column j to the left edge. The resulting matrix C is of the form in Step 1 or 2, and its $(1,1)$-minor C_{11} equals A_{ij}. Each row exchange, and (by Theorem 2.11) each column exchange, changes the sign of the determinant,

so

$$\det A = (-1)^{(i-1)+(j-1)} \det C = (-1)^{i+j} b \det C_{11} = (-1)^{i+j} b \det A_{ij}.$$

■

Theorem 2.13 (Cramer's rule)

Let A be an n by n matrix with successive columns $\mathbf{a}_1, \ldots, \mathbf{a}_n$. Let $\mathbf{x} = (x_1, \ldots, x_n)$ and $\mathbf{c} = (c_1, \ldots, c_n)$ be column matrices. If $A \cdot \mathbf{x} = \mathbf{c}$, then

$$(\det A) \cdot x_i = \det[\mathbf{a}_1 \cdots \mathbf{a}_{i-1}\ \mathbf{c}\ \mathbf{a}_{i+1} \cdots \mathbf{a}_n].$$

Proof. Let $C = [\mathbf{e}_1 \cdots \mathbf{e}_{i-1}\ \mathbf{x}\ \mathbf{e}_{i+1} \cdots \mathbf{e}_n]$. Since $A \cdot \mathbf{e}_j = \mathbf{a}_j$ and $A \cdot \mathbf{x} = \mathbf{c}$, we have $A \cdot C = [\mathbf{a}_1 \cdots \mathbf{a}_{i-1}\ \mathbf{c}\ \mathbf{a}_{i+1} \cdots \mathbf{a}_n]$, so by Theorem 2.10, $(\det A)(\det C)$ equals the right-hand side. C is I_n except for column i, which is $\mathbf{x}$; expanding by Lemma 2.12 along row i (whose only nonzero entry may be x_i in column i) gives $\det C = x_i(-1)^{i+i} \det I_{n-1} = x_i$. ■

Theorem 2.14

Let A be an n by n matrix of rank n; let $B = A^{-1}$. Then

$$b_{ij} = \frac{(-1)^{j+i} \det A_{ji}}{\det A}.$$

Proof. Fix j and let $\mathbf{x}$ be the jth column of B. Then $A \cdot B = I_n$ gives $A \cdot \mathbf{x} = \mathbf{e}_j$, and Cramer's rule gives $(\det A)x_i = \det[\mathbf{a}_1 \cdots \mathbf{a}_{i-1}\ \mathbf{e}_j\ \mathbf{a}_{i+1} \cdots \mathbf{a}_n]$. Column i of this matrix has a single nonzero entry 1 in row j, so by Lemma 2.12(b) the determinant is $1 \cdot (-1)^{j+i} \det A_{ji}$. Since $x_i = b_{ij}$, the theorem follows. ■

This yields the algorithm:

(1) form the **matrix of cofactors**, with (i, j) entry $(-1)^{i+j} \det A_{ij}$;

(2) transpose it;

(3) divide each entry by $\det A$. Its importance is theoretical (for instance it shows that the entries of A^{-1} are continuous functions of those of A); for computation, Gauss–Jordan reduction is more efficient.

2.5 Expansion by cofactors

Theorem 2.15

Let A be an n by n matrix and fix i. Then

$$\det A = \sum_{k=1}^{n} (-1)^{i+k} a_{ik} \det A_{ik}.$$

Proof. Let $A_i(\mathbf{x})$ be A with row i replaced by the row matrix $\mathbf{x}$. Row i of A is $\sum_k a_{ik}\mathbf{e}_k$, so by linearity (Axiom 2) and Lemma 2.12,

$$\det A = \sum_{k=1}^{n} a_{ik} \det A_i(\mathbf{e}_k) = \sum_{k=1}^{n} a_{ik}(-1)^{i+k} \det A_{ik}.$$

■

This is the expansion of the determinant by cofactors of the ith row; taking transposes gives expansion by cofactors of the jth column.

§3. Review of Topology in $\mathbb{R}^n$

3.1 Metric spaces

Definition 3.1 (metric space)

A **metric** on a set X is a function $d : X \times X \to \mathbb{R}$ such that for all $x, y, z \in X$:
(1) $d(x, y) = d(y, x)$;
(2) $d(x, y) \geq 0$, with equality iff $x = y$;
(3) $d(x, z) \leq d(x, y) + d(y, z)$. A **metric space** is a set with a specific metric. A subset $Y \subset X$ with the restricted metric is a **subspace** of X.

On $\mathbb{R}^n$ we have the **euclidean metric** $\|\mathbf{x} - \mathbf{y}\|$ and the **sup metric** $|\mathbf{x} - \mathbf{y}|$. $\mathbb{R}^n$ is called n-dimensional **euclidean space**.

Definition 3.2 (open and closed sets)

For $x_0 \in X$ and $\epsilon > 0$, the **ϵ-neighborhood** of x_0 is $U(x_0; \epsilon) = \{x : d(x, x_0) < \epsilon\}$. A set $U \subset X$ is **open** in X if for each $x_0 \in U$ some $U(x_0; \epsilon) \subset U$; a set C is **closed** in X if $X - C$ is open. Any open set containing x_0 is a **neighborhood** of x_0.

By the triangle inequality, an ϵ-neighborhood is open: if $d(x, x_0) < \epsilon$, then $U(x; \epsilon - d(x, x_0)) \subset U(x_0; \epsilon)$.

Theorem 3.1

Let (X, d) be a metric space. Then finite intersections and arbitrary unions of open sets of X are open in X. Similarly, finite unions and arbitrary intersections of closed sets of X are closed in X.

Proof (supplied). If x lies in a union of open sets, it lies in one of them, U_α, and some $U(x; \epsilon) \subset U_\alpha$ is contained in the union. If $x \in U_1 \cap \cdots \cap U_k$, choose ϵ_i with $U(x; \epsilon_i) \subset U_i$ and let $\epsilon = \min \epsilon_i > 0$. The statements about closed sets follow by De Morgan's laws. ■

Theorem 3.2

Let X be a metric space and Y a subspace. A subset A of Y is open in Y if and only if $A = U \cap Y$ for some U open in X. Similarly, A is closed in Y if and only if $A = C \cap Y$ for some C closed in X.

Proof (supplied). The ϵ-neighborhood of $y \in Y$ in Y is $U_X(y; \epsilon) \cap Y$. If U is open in X and $y \in U \cap Y$, then $U_X(y; \epsilon) \subset U$ for some ϵ, so $U_Y(y; \epsilon) \subset U \cap Y$. Conversely, if A is open in Y, choose for each $y \in A$ an ϵ_y with $U_X(y; \epsilon_y) \cap Y \subset A$ and put $U = \bigcup_{y \in A} U_X(y; \epsilon_y)$; then U is open in X and $U \cap Y = A$. For closed sets take complements: $Y - A = (X - C) \cap Y$. ■

Hence if A is open in Y and Y is open in X, then A is open in X; likewise for closed sets.

Definition 3.3 (limit point, closure)

$x_0 \in X$ is a **limit point** of $A \subset X$ if every ϵ-neighborhood of x_0 meets A in a point other than x_0 (equivalently, in infinitely many points of A). The **closure** $\bar{A}$ of A is A together with all its limit points.

Theorem 3.3

If A is a subset of X, then $\bar{A}$ is a closed set of X. A subset of X is closed if and only if it contains all its limit points.

Proof (supplied). A set C is closed iff every $x \notin C$ has a neighborhood disjoint from C, i.e. iff no point outside C is a limit point of C; this is the second statement. For the first: if $x \notin \bar{A}$, some $U(x;\epsilon)$ misses A. This open set also misses every limit point y of A: a point $y \in U(x;\epsilon)$ has a neighborhood inside $U(x;\epsilon)$, which contains no point of A. So $U(x;\epsilon) \cap \bar{A} = \emptyset$ and $X - \bar{A}$ is open. ■

In $\mathbb{R}^n$, the ϵ-neighborhood of $\mathbf{a}$ in the euclidean metric is the **open ball** $B(\mathbf{a};\epsilon)$; in the sup metric it is the **open cube** $C(\mathbf{a};\epsilon)$. By (1.1),

$$B(\mathbf{a};\epsilon) \subset C(\mathbf{a};\epsilon) \subset B(\mathbf{a};\sqrt{n}\,\epsilon).$$

Theorem 3.4

If X is a subspace of $\mathbb{R}^n$, the collection of open sets of X is the same whether one uses the euclidean metric or the sup metric on X. The same is true for the collection of closed sets of X.

Proof (supplied). If A is open for the sup metric and $\mathbf{x} \in A$, then $C(\mathbf{x};\epsilon) \cap X \subset A$ for some ϵ, hence $B(\mathbf{x};\epsilon) \cap X \subset A$. If A is open for the euclidean metric, $B(\mathbf{x};\epsilon) \cap X \subset A$, hence $C(\mathbf{x};\epsilon/\sqrt{n}) \cap X \subset A$. Closed sets are the complements. ■

A property of X depending only on its collection of open sets is a **topological property**; limits, continuity and compactness are examples.

3.2 Limits and continuity

Definition 3.4 (continuity)

Let X, Y be metric spaces. $f : X \to Y$ is **continuous at** x_0 if for each open $V \ni f(x_0)$ there is an open $U \ni x_0$ with $f(U) \subset V$; equivalently, for each $\epsilon > 0$ there is $\delta > 0$ with $d_Y(f(x), f(x_0)) < \epsilon$ whenever $d_X(x, x_0) < \delta$. f is **continuous** if it is continuous at every point; equivalently, $f^{-1}(V)$ is open in X for each open V of Y (or $f^{-1}(D)$ is closed for each closed D).

If some δ-neighborhood of x_0 is $\{x_0\}$, then x_0 is an **isolated point** of X, and every f is continuous at x_0. Constant maps and the identity are continuous.

Theorem 3.5

(a) Let $x_0 \in A$, where A is a subspace of X. If $f : X \to Y$ is continuous at x_0, then $f|A : A \to Y$ is continuous at x_0.
(b) Let $f : X \to Y$ and $g : Y \to Z$. If f is continuous at x_0 and g is continuous at $y_0 = f(x_0)$, then $g \circ f : X \to Z$ is continuous at x_0.

Proof (supplied). (a) The same δ works for points of A.

(b) Given ϵ, choose η with $d_Z(g(y), g(y_0)) < \epsilon$ for $d_Y(y, y_0) < \eta$, then δ with $d_Y(f(x), y_0) < \eta$ for $d_X(x, x_0) < \delta$. ■

Theorem 3.6

(a) Let X be a metric space and $f : X \to \mathbb{R}^n$, $f(x) = (f_1(x), \dots, f_n(x))$. Then f is continuous at x_0 if and only if each **component function** $f_i : X \to \mathbb{R}$ is continuous at x_0.
(b) Let $f, g : X \to \mathbb{R}$ be continuous at x_0. Then $f+g$, $f-g$ and $f \cdot g$ are continuous at x_0, and f/g is continuous at x_0 if $g(x_0) \neq 0$.
(c) The projection $\pi_i : \mathbb{R}^n \to \mathbb{R}$, $\pi_i(\mathbf{x}) = x_i$, is continuous.

Proof (supplied). (a) Use the sup metric: $|f(x) - f(x_0)| < \epsilon$ iff $|f_i(x) - f_i(x_0)| < \epsilon$ for all i; take the minimum of the n values of δ.

(c) $|\pi_i(\mathbf{x}) - \pi_i(\mathbf{y})| \leq |\mathbf{x} - \mathbf{y}|$.

(b) Addition, subtraction and multiplication $\mathbb{R}^2 \to \mathbb{R}$ are continuous, and so is division on $\{(s,t) : t \neq 0\}$, by the usual estimates, e.g. $|st - s_0t_0| \leq |s||t - t_0| + |t_0||s - s_0|$. Compose with $x \mapsto (f(x), g(x))$, continuous by (a), and use Theorem 3.5. ∎

Thus functions built from the familiar continuous functions by algebraic operations and composition are continuous, e.g. $f(s,t,u,v) = \sin(s+t)/e^{uv}$ on $\mathbb{R}^4$.

Definition 3.5 (limit)

Let $A \subset X$, $f : A \to Y$, and let x_0 be a limit point of A (not necessarily in A). We say $f(x) \to y_0$ as $x \to x_0$, or $\lim_{x \to x_0} f(x) = y_0$, if for each open $V \ni y_0$ there is an open $U \ni x_0$ with $f(x) \in V$ for all $x \in U \cap A$, $x \neq x_0$. Equivalently: for each $\epsilon > 0$ there is $\delta > 0$ with $d_Y(f(x), y_0) < \epsilon$ whenever $x \in A$ and $0 < d_X(x, x_0) < \delta$.

We do not attempt to define the limit if x_0 is not a limit point of the domain; the value of f at x_0 plays no role.

Theorem 3.7

Let $f : X \to Y$. If x_0 is an isolated point of X, then f is continuous at x_0. Otherwise, f is continuous at x_0 if and only if $f(x) \to f(x_0)$ as $x \to x_0$.

Proof (supplied). Compare the ϵ-δ conditions: they differ only in whether $x = x_0$ is allowed, and at $x = x_0$ the condition $d_Y(f(x), f(x_0)) < \epsilon$ holds trivially. ∎

Theorem 3.8

(a) Let $A \subset X$ and $f : A \to \mathbb{R}^n$, $f = (f_1, \dots, f_n)$; let $\mathbf{a} = (a_1, \dots, a_n)$. Then $f(x) \to \mathbf{a}$ as $x \to x_0$ if and only if $f_i(x) \to a_i$ for each i.
(b) Let $f, g : A \to \mathbb{R}$. If $f(x) \to a$ and $g(x) \to b$ as $x \to x_0$, then $f+g \to a+b$, $f-g \to a-b$, $f \cdot g \to a \cdot b$, and $f/g \to a/b$ if $b \neq 0$.

Proof (supplied). Extend f, g to $A \cup \{x_0\}$ by $f(x_0) = \mathbf{a}$ (resp. a, b). The limit statements become continuity at x_0 of the extended functions (Theorem 3.7), and Theorem 3.6 applies. ∎

3.3 Interior and exterior

Definition 3.6 (interior, exterior, boundary)

For $A \subset \mathbb{R}^n$: $\operatorname{Int} A$ is the union of all open sets of $\mathbb{R}^n$ contained in A; $\operatorname{Ext} A$ is the union of all open sets disjoint from A; $\operatorname{Bd} A$ consists of the points in neither $\operatorname{Int} A$ nor $\operatorname{Ext} A$.

A point $\mathbf{x}$ is in $\operatorname{Bd} A$ iff every open set containing $\mathbf{x}$ meets both A and $\mathbb{R}^n - A$. $\mathbb{R}^n$ is the disjoint union of $\operatorname{Int} A$, $\operatorname{Ext} A$ and $\operatorname{Bd} A$; the first two are open and $\operatorname{Bd} A$ is closed.

For a **rectangle** $Q = [a_1, b_1] \times \cdots \times [a_n, b_n]$ one checks $\operatorname{Int} Q = (a_1, b_1) \times \cdots \times (a_n, b_n)$ (an **open rectangle**), $\operatorname{Ext} Q = \mathbb{R}^n - Q$, $\operatorname{Bd} Q = Q - \operatorname{Int} Q$. The open cube $C(\mathbf{a}; \epsilon) = (a_1 - \epsilon, a_1 + \epsilon) \times \cdots \times (a_n - \epsilon, a_n + \epsilon)$ is an open rectangle; the corresponding closed rectangle $[a_1 - \epsilon, a_1 + \epsilon] \times \cdots \times [a_n - \epsilon, a_n + \epsilon]$ is the **closed cube** (or simply cube) centered at $\mathbf{a}$.

§4. Compact Subspaces and Connected Subspaces of $\mathbb{R}^n$

4.1 Compact spaces

Definition 4.1 (compact)

Let $X \subset \mathbb{R}^n$. A **covering** of X is a collection of subsets of $\mathbb{R}^n$ whose union contains X; it is an **open covering** if each member is open in $\mathbb{R}^n$. X is **compact** if every open covering of X contains a finite subcollection that also covers X.

Theorem 4.1

A subspace X of $\mathbb{R}^n$ is compact if and only if for every collection of sets open in X whose union is X, there is a finite subcollection whose union equals X.

Proof. Suppose X is compact and $\{A_\alpha\}$ are open in X with union X. By Theorem 3.2 choose U_α open in $\mathbb{R}^n$ with $A_\alpha = U_\alpha \cap X$. The U_α cover X, so finitely many, $U_{\alpha_1}, \ldots, U_{\alpha_k}$, do; then $A_{\alpha_1}, \ldots, A_{\alpha_k}$ have union X. Conversely, given an open covering $\{U_\alpha\}$ of X by open sets of $\mathbb{R}^n$, the sets $U_\alpha \cap X$ are open in X with union X; a finite subfamily with union X gives finitely many U_α covering X. ■

Theorem 4.2

The subspace $[a, b]$ of $\mathbb{R}$ is compact.

Proof (supplied). Let $\mathcal{A}$ be an open covering of $[a, b]$ and let S be the set of $x \in [a, b]$ such that $[a, x]$ is covered by finitely many members of $\mathcal{A}$. Then $a \in S$; let $c = \sup S$. Some $U \in \mathcal{A}$ contains c, and with it an interval $(c - \delta, c + \delta)$. There is $x \in S$ with $x > c - \delta$; adding U to the finite family covering $[a, x]$ covers $[a, \min(c + \delta/2, b)]$. Hence $c \in S$, and if $c < b$ we would get a point of S larger than c. So $c = b \in S$. ■

Definition 4.2 (bounded)

$X \subset \mathbb{R}^n$ is **bounded** if there is M with $|\mathbf{x}| \leq M$ for all $\mathbf{x} \in X$.

Theorem 4.3

If X is a compact subspace of $\mathbb{R}^n$, then X is closed and bounded.

Proof.

Step 1. Bounded: the open cubes $U_N = C(\mathbf{0}; N)$, $N = 1, 2, \ldots$, are open, increasing, and cover $\mathbb{R}^n$, hence X. Finitely many, $U_{N_1}, \ldots, U_{N_k}$, cover X; with $M = \max N_i$, $X \subset U_M$.

Step 2. Closed: let $\mathbf{a} \notin X$; we find a neighborhood of $\mathbf{a}$ disjoint from X. Let $C_N = \{\mathbf{x} : |\mathbf{x} - \mathbf{a}| \leq 1/N\}$ and $V_N = \mathbb{R}^n - C_N$. The V_N are open, increasing, and cover $\mathbb{R}^n - \{\mathbf{a}\} \supset X$. Finitely many

cover X, so $X \subset V_M$ for some M; then C_M is disjoint from X, and in particular the open cube $C(\mathbf{a}; 1/M)$ lies in $\mathbb{R}^n - X$. ■

Corollary 4.4

Let X be a compact subspace of $\mathbb{R}$. Then X has a largest element and a smallest element.

Proof. X is bounded, so it has a supremum and an infimum; X is closed, so these (being limit points of X or points of X) belong to X. ■

Theorem 4.5 (Extreme-value theorem)

Let X be a compact subspace of $\mathbb{R}^m$. If $f : X \to \mathbb{R}^n$ is continuous, then $f(X)$ is a compact subspace of $\mathbb{R}^n$. In particular, if $\phi : X \to \mathbb{R}$ is continuous, then ϕ has a maximum value and a minimum value.

Proof. Let $\{V_\alpha\}$ be an open covering of $f(X)$. The sets $f^{-1}(V_\alpha)$ are open in X and cover X, so by Theorem 4.1 finitely many, for $\alpha = \alpha_1, \ldots, \alpha_k$, cover X; then $V_{\alpha_1}, \ldots, V_{\alpha_k}$ cover $f(X)$. If $\phi : X \to \mathbb{R}$ is continuous, $\phi(X)$ is compact, and by Corollary 4.4 it has a largest and a smallest element. ■

Definition 4.3 (ϵ-neighborhood of a set)

For $X \subset \mathbb{R}^n$ and $\epsilon > 0$, the union of the balls $B(\mathbf{a}; \epsilon)$, $\mathbf{a} \in X$, is the **ϵ-neighborhood** of X in the euclidean metric; the union of the cubes $C(\mathbf{a}; \epsilon)$ is the ϵ-neighborhood in the sup metric.

Theorem 4.6 (The ϵ-neighborhood theorem)

Let X be a compact subspace of $\mathbb{R}^n$; let U be an open set of $\mathbb{R}^n$ containing X. Then there is an $\epsilon > 0$ such that the ϵ-neighborhood of X (in either metric) is contained in U.

Proof. The euclidean ϵ-neighborhood lies inside the sup-metric one, so it suffices to treat the sup metric.

Step 1. For $C \subset \mathbb{R}^n$ define $d(\mathbf{x}, C) = \inf\{|\mathbf{x} - \mathbf{c}| : \mathbf{c} \in C\}$, the **distance from $\mathbf{x}$ to C**. It is continuous: for $\mathbf{c} \in C$, $d(\mathbf{x}, C) - |\mathbf{x} - \mathbf{y}| \le |\mathbf{x} - \mathbf{c}| - |\mathbf{x} - \mathbf{y}| \le |\mathbf{y} - \mathbf{c}|$; taking the infimum over $\mathbf{c}$, $d(\mathbf{x}, C) - d(\mathbf{y}, C) \le |\mathbf{x} - \mathbf{y}|$, and symmetrically.

Step 2. Let $f(\mathbf{x}) = d(\mathbf{x}, \mathbb{R}^n - U)$ for $\mathbf{x} \in X$. Then f is continuous and positive on X: if $\mathbf{x} \in X$, some $C(\mathbf{x}; \delta) \subset U$, so $f(\mathbf{x}) \ge \delta$. Since X is compact, f has a minimum value $\epsilon > 0$ (Theorem 4.5). Then for $\mathbf{a} \in X$, every point $\mathbf{y}$ with $|\mathbf{y} - \mathbf{a}| < \epsilon$ lies in U; i.e. the ϵ-neighborhood of X lies in U. ■

Some hypothesis on X is needed: if X is the x-axis in $\mathbb{R}^2$ and $U = \{(x, y) : y^2 < 1/(1 + x^2)\}$, no ϵ-neighborhood of X lies in U.

Theorem 4.7 (Uniform continuity)

Let X be a compact subspace of $\mathbb{R}^m$; let $f : X \to \mathbb{R}^n$ be continuous. Given $\epsilon > 0$, there is a $\delta > 0$ such that whenever $\mathbf{x}, \mathbf{y} \in X$,

$$|\mathbf{x} - \mathbf{y}| < \delta \quad \text{implies} \quad |f(\mathbf{x}) - f(\mathbf{y})| < \epsilon.$$

This result also holds if one uses the euclidean metric instead of the sup metric.

The conclusion is the condition of **uniform continuity**.

Proof. Consider $X \times X \subset \mathbb{R}^m \times \mathbb{R}^m$ and the **diagonal** $\Delta = \{(\mathbf{x}, \mathbf{x}) : \mathbf{x} \in X\}$, which is compact as the image of X under the continuous map $\mathbf{x} \mapsto (\mathbf{x}, \mathbf{x})$. First use the euclidean metric. The function $g : X \times X \to \mathbb{R}$, $g(\mathbf{x}, \mathbf{y}) = \|f(\mathbf{x}) - f(\mathbf{y})\|$, is continuous, so $\{(\mathbf{x}, \mathbf{y}) : g(\mathbf{x}, \mathbf{y}) < \epsilon\}$ is open

in $X \times X$ and contains Δ; hence it equals $U \cap (X \times X)$ for some open $U \subset \mathbb{R}^{2m}$ containing Δ (Theorem 3.2). By Theorem 4.6, some δ-neighborhood of Δ lies in U. If $\mathbf{x}, \mathbf{y} \in X$ with $\|\mathbf{x}-\mathbf{y}\| < \delta$, then $\|(\mathbf{x},\mathbf{y})-(\mathbf{y},\mathbf{y})\| = \|(\mathbf{x}-\mathbf{y},\mathbf{0})\| = \|\mathbf{x}-\mathbf{y}\| < \delta$, so $(\mathbf{x},\mathbf{y})$ lies in the δ-neighborhood of Δ, hence in U, and $g(\mathbf{x},\mathbf{y}) < \epsilon$.

For the sup metric: if $|\mathbf{x}-\mathbf{y}| < \delta/\sqrt{m}$ then $\|\mathbf{x}-\mathbf{y}\| < \delta$, whence $|f(\mathbf{x})-f(\mathbf{y})| \le \|f(\mathbf{x})-f(\mathbf{y})\| < \epsilon$. ■

Lemma 4.8

The rectangle $Q = [a_1, b_1] \times \cdots \times [a_n, b_n]$ in $\mathbb{R}^n$ is compact.

Proof. Induction on n; the case $n = 1$ is Theorem 4.2. Write $Q = X \times [a_n, b_n]$ with X a rectangle in $\mathbb{R}^{n-1}$, compact by the induction hypothesis. Let $\mathcal{A}$ be an open covering of Q.

Step 1. Given $t \in [a_n, b_n]$, there is $\epsilon > 0$ such that $X \times (t-\epsilon, t+\epsilon)$ is covered by finitely many elements of $\mathcal{A}$. Indeed, $X \times t$ is compact (the image of X under $\mathbf{x} \mapsto (\mathbf{x}, t)$), so it is covered by finitely many $A_1, \ldots, A_k \in \mathcal{A}$. Their union U is open and contains $X \times t$; by Theorem 4.6 some ϵ-neighborhood of $X \times t$ lies in U, and in particular $X \times (t-\epsilon, t+\epsilon) \subset U$.

Step 2. By Step 1 choose for each t an open interval $V_t \ni t$ such that $X \times V_t$ is covered by finitely many elements of $\mathcal{A}$. The V_t cover $[a_n, b_n]$, so finitely many, $V_{t_1}, \ldots, V_{t_m}$, do. Then Q is contained in $\bigcup_i X \times V_{t_i}$, each covered by finitely many elements of $\mathcal{A}$; so Q is covered by finitely many. ■

Theorem 4.9

If X is a closed and bounded subspace of $\mathbb{R}^n$, then X is compact.

Proof. Let $\mathcal{A}$ be an open covering of X. Adjoin the open set $\mathbb{R}^n - X$ to get an open covering of $\mathbb{R}^n$. Since X is bounded, it lies in a rectangle Q, which is compact (Lemma 4.8); finitely many members of the enlarged collection cover Q. Discarding $\mathbb{R}^n - X$ if it occurs leaves finitely many members of $\mathcal{A}$; they may not cover Q, but they cover X, since $\mathbb{R}^n - X$ contains no point of X. ■

All theorems of this section hold for arbitrary metric spaces in place of $\mathbb{R}^n, \mathbb{R}^m$, *except* Theorem 4.9.

4.2 Connected spaces

Definition 4.4 (connected)

A metric space X is **connected** if X cannot be written as the union of two disjoint non-empty sets A and B, each of which is open in X.

Theorem 4.10

The closed interval $[a, b]$ of $\mathbb{R}$ is connected.

Proof (supplied). Suppose $[a,b] = A \cup B$ with A, B disjoint, non-empty and open in $[a,b]$ (hence also closed in $[a,b]$); say $a \in A$. Let $c = \sup\{x \in [a,b] : [a,x] \subset A\}$. Since A is closed, $c \in A$. If $c < b$, openness of A gives $[c, c+\delta) \subset A$ for some $\delta > 0$, so $[a, c+\delta/2] \subset A$, contradicting the definition of c. So $c = b$ and $[a,b] \subset A$, so $B = \emptyset$, a contradiction. ■

Theorem 4.11 (Intermediate-value theorem)

Let X be connected. If $f : X \to Y$ is continuous, then $f(X)$ is a connected subspace of Y. In particular, if $\phi : X \to \mathbb{R}$ is continuous and $f(x_0) < r < f(x_1)$ for some points x_0, x_1 of X, then $f(x) = r$ for some point x of X.

Proof. If $f(X) = A \cup B$ with A, B disjoint and open in $f(X)$, then $f^{-1}(A)$ and $f^{-1}(B)$ are disjoint, open in X, with union X; connectedness of X forces one of them, hence one of A, B, to be empty. For ϕ: let $A = \{y \in \mathbb{R} : y < r\}$, $B = \{y : y > r\}$, open in $\mathbb{R}$. If $r \notin \phi(X)$, then $\phi(X)$ is the union of the disjoint sets $\phi(X) \cap A$, $\phi(X) \cap B$, open in $\phi(X)$ and both non-empty, contradicting connectedness of $\phi(X)$. ■

Definition 4.5 (line segment, convex)

The **line segment** joining $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ is the set of points $\mathbf{x} = \mathbf{a} + t(\mathbf{b} - \mathbf{a})$, $0 \leq t \leq 1$. A set $A \subset \mathbb{R}^n$ is **convex** if for every pair $\mathbf{a}, \mathbf{b} \in A$ the segment joining them lies in A.

A line segment is connected, being the image of $[0, 1]$ under $t \mapsto \mathbf{a} + t(\mathbf{b} - \mathbf{a})$. Every convex set A is connected: if $A = U \cup V$ with U, V disjoint, non-empty and open in A, pick $\mathbf{a} \in U$, $\mathbf{b} \in V$ and let L be the segment joining them; then $U \cap L$ and $V \cap L$ are disjoint, non-empty, open in L, with union L, a contradiction. Hence open balls, open cubes and rectangles in $\mathbb{R}^n$ are connected.